6. Automated and emerging technologies

6.2 Robotics

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

1 A company uses robots in its factory to manufacture large pieces of furniture.

(a) One characteristic of a robot is that it is programmable.

State **two** other characteristics of a robot.

1 .....

.....

2 .....

..... [2]

(b) Give **two** advantages to company employees of using robots to manufacture large pieces of furniture.

1 .....

.....

2 .....

..... [2]

(c) Give **one** disadvantage to the company's owners of using robots to manufacture large pieces of furniture.

.....

..... [1]

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

2 A farmer uses an automated robot to plant seeds in the field.

(a) State what is meant by the robot being automated.

.....

..... [1]

(b) Give **three** characteristics of a robot.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

(d) Give **two** advantages of the farmer using an automated robot to plant seeds.

1 .....

.....

.....

2 .....

.....

.....

[2]

(e) Give **two** disadvantages of the farmer using an automated robot to plant seeds.

1 .....

.....

.....

2 .....

.....

.....

[2]

(f) The robot is adapted to have machine learning capabilities.

Explain how this will improve the robot.

.....

.....

.....

..... [2]

SHOWMIK SIR Key to Computer Science

Musfiqul Hamid Showmik
01797806089

6. Automated and emerging technologies

6.3 Artificial intelligence

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

1 One component of an expert system is the inference engine.

(a) Identify the **three** other components in an expert system.

1 .....

2 .....

3 ..... [3]

(b) Describe the role of the inference engine in an expert system.

.....

.....

.....

..... [2]

2 The rule base and the inference engine are two components of an expert system.

(a) Identify the other **two** components of an expert system.

1 .....

2 ..... [2]

(b) Describe the role of the rule base in an expert system.

.....

.....

.....

..... [2]

SHOWMIKSIR Key to Computer Science

Musfiqul Hamid Showmik
01797806089

3 A business has a system that is described as having artificial intelligence (AI).

(a) State **one** of the main characteristics of an AI system.

.....

..... [1]

(b) An AI system is an expert system.

Explain how an expert system operates.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [6]

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

4 A car repair garage uses an expert system.

(a) Complete the description about the operation of the expert system.

Use the terms from the list. Some of the terms in the list will **not** be used.

inference engine interface knowledge base

machine learning mechanical engine output device

question base rule base

An expert system has a ..... that contains a list of facts.

The ..... applies the .....

to the ..... to reach a diagnosis for the repair of the car.

The user provides data to the system using an ..... .

[5]

(b) The expert system has machine learning capabilities.

Describe what is meant by machine learning capabilities.

.....

.....

.....

.....

.....

.....

.....

..... [4]

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

5 A garage uses an expert system to help diagnose any problems with cars that need repair.

(a) The expert system is an example of artificial intelligence (AI).

Describe what is meant by AI.

.....

.....

.....

..... [2]

(b) A car has a problem with its braking system, so it is brought into the garage.

Explain how the expert system operates and how it is used to help diagnose the problem.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089

6 A doctor's surgery has an expert system that helps diagnose the illnesses of its patients.

(a) Complete the paragraph about the operation of the expert system.

Use the terms from the list.

Some of the terms in the list will **not** be used. Some terms may be used more than once.

artificial binary central processing unit (CPU) data

inference engine interface knowledge base primary storage

process real read only memory (ROM) rule base

An expert system is a type of ..... intelligence. The doctor will type data about the symptoms of the illness into the ..... . The ..... will decide which questions to ask the doctor. It will do this by linking the facts in the ..... to the ..... .

The ..... will decide on a diagnosis and this will be output on the ..... .

[7]

(b) The expert system has the ability to automatically adapt its own processes and data.

Give the name of this ability.

..... [1]

Musfiqul Hamid Showmik
01797806089

7 **(b)** An expert system can be used to diagnose an error with a robot vacuum cleaner.

Circle **three** components that are part of an expert system.

knowledge base operating system firewall

server printer actuator inference engine

rule base encryption

[3]

(c) Explain how the robot vacuum cleaner can make use of machine learning.

.....

.....

.....

.....

.....

..... [3]

8 **(d)** The plough uses artificial intelligence (AI) to navigate its way around the field.

Explain how the plough makes use of AI for this purpose.

.....

.....

.....

.....

.....

.....

.....

..... [4]

SHOWMIK SIR
Key to Computer Science

Musfiqul Hamid Showmik
01797806089